

Paper Folding & Cutting

COMPLETE LEARNING GUIDE

What You'll Learn

Paper folding and cutting questions test spatial visualization of what happens when a hole is punched or a shape is cut into a sheet of paper that has been folded one or more times, and the sheet is then unfolded flat again. Every question in this family reduces to one idea: each fold line acts as a mirror, and any hole or cut made through the folded layers is reflected across every fold line it crosses when the paper is opened back out. The number of times a mark is duplicated depends on how many layers of paper the punch or cut passed through, which in turn depends on how many folds separate that point from the sheet's original single-layer form. This sub-module builds the fold-as-mirror method from a single fold up through double (quarter) folds and diagonal folds, covers the special case of a hole placed exactly on a crease, extends the same logic to cuts rather than punched holes, and shows how to count the resulting number of holes without redrawing the whole sheet.

1. The Core Principle: Every Fold Is a Mirror Line

Treat each fold as a mirror placed along the crease: whatever is punched or cut through the folded stack gets reflected across that crease when the paper reopens. If a sheet is folded once and a hole is punched anywhere except exactly on the fold, the hole exists in 2 layers of paper, so unfolding produces 2 holes that are mirror images of each other across the fold line — same distance from the crease, on opposite sides.

- A rectangular sheet is folded in half along its vertical midline (left edge onto right edge). A hole is punched 3 cm from the folded edge (the crease), 2 cm down from the top. Unfolding gives 2 holes, both 2 cm from the top, one 3 cm to the left of the original center line and one 3 cm to the right — a mirror pair across the vertical midline.

2. Double (Quarter) Folds and the Four-Copy Rule

Folding a sheet twice — once vertically, once horizontally — produces 4 layers, so a single hole punched through all 4 layers at a point away from both creases reflects across both fold lines and appears 4 times when fully unfolded, positioned symmetrically in all four quadrants of the original sheet.

- A square sheet is folded in half top-to-bottom, then in half again left-to-right, leaving a small square packet with the original center at one corner. A hole is punched at the OUTER corner of this packet (the corner diagonally farthest from the original center). Unfolding reveals 4 holes, one near each of the original square's 4 corners, symmetric about both the vertical and horizontal midlines.

3. The Special Case: A Hole Placed Exactly on a Fold Line

If a hole lands exactly ON a crease rather than away from it, that crease can't duplicate it (there's nothing on the other side of that particular fold to reflect it into, since the hole already straddles the line) — so a hole on one fold line, in a double-fold packet, produces only 2 copies instead of 4.

- Same double-fold packet as above (folded top-to-bottom, then left-to-right). This time the hole is punched exactly on the crease that was the vertical fold, but away from the horizontal crease. Unfolding gives only 2 holes (not 4), both lying exactly on the original sheet's vertical midline, mirrored top-to-bottom across the horizontal fold.

4. Cutting a Notch from a Folded Edge

Cutting (rather than punching) a small triangular or semicircular notch out of the folded edge itself removes material from both layers at once along that edge — when unfolded, the two mirrored notch-halves join into a single complete symmetric shape straddling the original fold line, rather than two separate holes.

- A square sheet is folded in half (left onto right). A small triangular notch is cut out of the middle of the folded (creased) edge, with the triangle's flat side lying along the crease. Unfolding shows one complete diamond (rhombus)-shaped hole centered exactly on the sheet's original vertical midline — the two mirrored half-triangles have joined into one whole shape.

5. Multiple Holes in One Folded Packet

When more than one hole is punched into the same folded packet, apply the mirroring rule to each hole independently and then combine the results — every hole away from all creases multiplies by 2 for each fold it survives, and the total hole count is the sum of each hole's individual multiplication.

- A sheet folded twice (4 layers) has TWO holes punched at different points, both away from every crease. Each hole individually produces 4 copies on unfolding (per the four-copy rule), so the total is $4 + 4 = 8$ holes scattered across the sheet in two sets of 4 mirror-symmetric points.

6. Diagonal Folds

When a square is folded along its diagonal (corner to corner) instead of along the midline, the crease itself is the diagonal line, and the same mirroring principle applies — but the reflection axis is now that diagonal rather than a horizontal or vertical line, so a hole's mirrored position must be measured perpendicular to the diagonal, not perpendicular to an edge.

- A square sheet is folded once along its main diagonal (bringing one corner onto the opposite corner) to form a triangle. A hole is punched close to the folded (hypotenuse) edge but off it. Unfolding gives 2 holes, symmetric about the diagonal line — both equally close to the diagonal but on opposite sides of it, not opposite sides of a horizontal or vertical line.

7. Unfolding in Reverse Order

When a sheet has been folded in a specific sequence (fold 1, then fold 2, then fold 3), mentally unfold it in the exact REVERSE order (undo fold 3 first, then fold 2, then fold 1) — reflecting across the most recent crease first, since that crease is the outermost layer of the folded packet and must be opened before the earlier creases become accessible.

- A rectangle is folded left-to-right (fold 1), then the resulting shape is folded top-to-bottom (fold 2), and a hole is punched. To find the unfolded pattern, first undo fold 2 (reflect the hole top-to-bottom to get 2 holes), THEN undo fold 1 (reflect both of those left-to-right to get 4 holes total) — undoing in the wrong order would place the intermediate reflection relative to the wrong edge.

Common Pitfalls

- Forgetting that a hole placed exactly ON a fold line produces half as many copies as a hole placed away from every crease.
- Applying the four-copy rule to every hole regardless of position, without checking whether the hole actually lies on one of the two fold lines.
- Treating a cut made across a folded edge as if it produces two separate holes instead of one joined symmetric shape.
- Mirroring a hole's position perpendicular to a sheet edge when the actual fold was diagonal, ignoring that the true mirror axis is the diagonal crease.
- Unfolding the creases in the same order they were folded instead of the reverse order.
- Losing track of which corner or edge of the small folded packet corresponds to which corner or edge of the original full sheet.
- Assuming the number of holes always equals $2 \times$ the number of folds, rather than $2^{(\text{number of folds})}$ for a hole away from every crease.

Key Takeaways

- Every fold is a mirror line: a hole or cut away from all creases doubles for each fold it passes through, giving $2^{(\text{number of folds})}$ copies.
- A hole placed exactly on one fold line is only reflected by the OTHER folds, so it produces half as many copies as usual.
- A cut made across a folded edge joins into one complete symmetric shape when unfolded, not separate mirrored holes.
- Diagonal folds mirror perpendicular to the diagonal crease, not perpendicular to a horizontal or vertical edge.
- Always unfold mentally in reverse order — undo the last fold made first, then work backward to the first fold.

- Sketch or label each crease as you go so you don't lose track of which small-packet corner maps to which original-sheet corner.

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